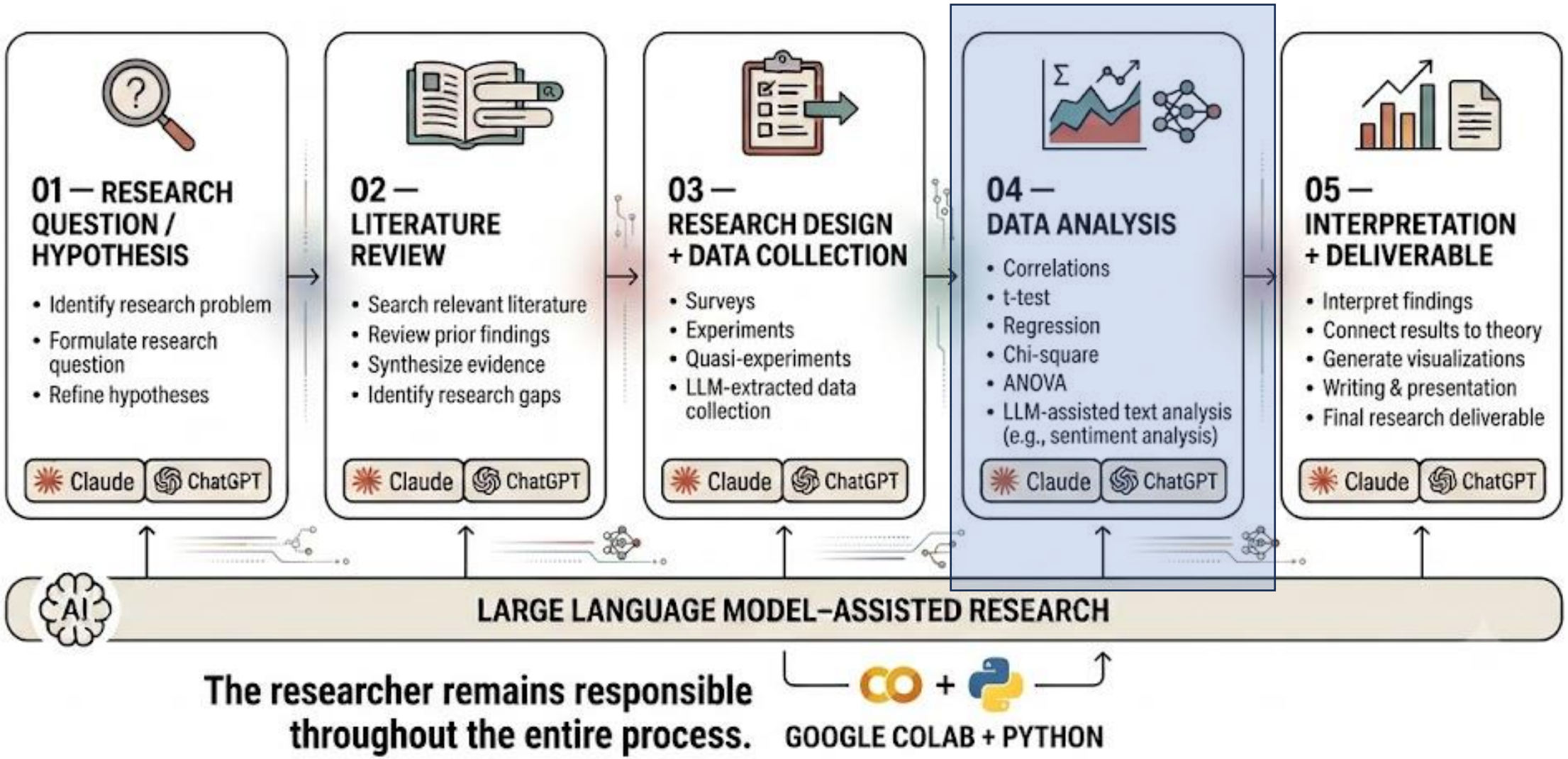
Sentiment Classification

JOUR 701

Research Methods in the Age of AI



The researcher remains responsible throughout the entire process.

GOOGLE COLAB + PYTHON

What is computational text analysis?

NEW CONCEPT

01 A family of methods for examining patterns in written or spoken language using computers.

02 Inputs: survey responses, comments, headlines, transcripts, or articles.

03 Outputs: counts, indicators, categories, scores, or patterns that researchers interpret.

Text → documented transformation → measure → interpretation.

Recap: Tuesday's methods

COMPARE

What we did

Word count → length
Word frequency → common terms
Keyword match → mentions of trust

What we could not conclude

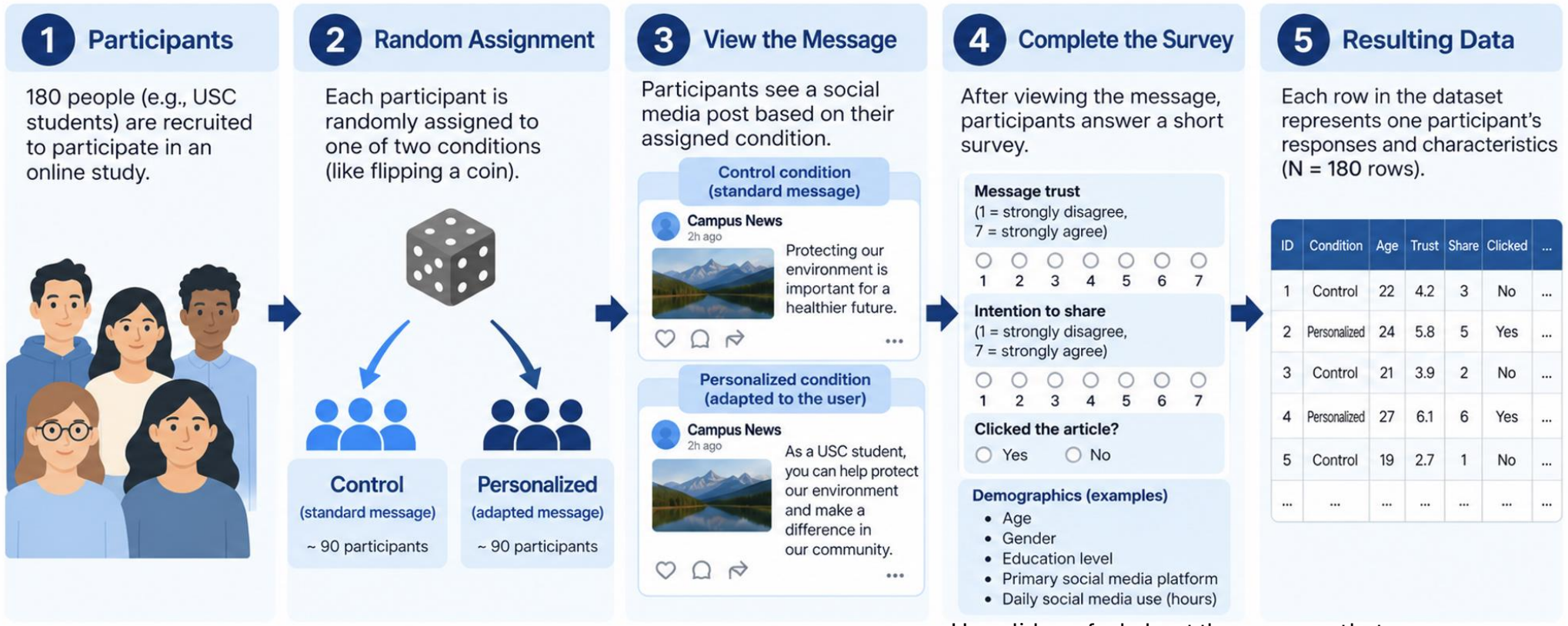
A word count does not measure emotion.

"Trust" can appear in supportive and critical responses.

Today: from words to carefully evaluated sentiment labels.

Same dataset

A randomized experiment with two conditions and a post-message survey (N = 180)



How did you feel about the message that you just read?

Today we use this SAME dataset to ask three different statistical questions.

Remember our study

X = CONDITION

Control vs. Personalized message (randomly assigned).

TEXT OUTCOME

Open response: “How did you feel about the message?”

RESEARCH QUESTION

Does response sentiment vary across the two conditions?

What is sentiment classification?

WORKED EXAMPLE

“The wording was clear and informative.”



A system assigns a label, such as POSITIVE.

Sentiment is an operationalized interpretation—not directly observed emotion.

Two approaches to sentiment analysis

METHOD COMPARISON

1 Lexicon-based

Example: VADER

Uses a dictionary of words with sentiment scores and rules.

“Not good” → rules account for negation.

Strength: transparent and fast.

Limitation: can miss sarcasm or nuance.

2 LLM-based

Example: prompt an LLM

Uses a trained language model to interpret text in context.

Prompt: “Classify this response as positive, negative, or neutral.”

Strength: can capture context and nuance.

Limitation: predictions can be wrong or vary.

Today: we focus on the lexicon-based approach to understand how sentiment scores are created.

How was the VADER lexicon created?

From human judgments to a reusable sentiment tool

1

Collect words

great • awful
lol • :)

→

2

Human ratings

−4 to +4
negative → positive

→

3

Build lexicon

Average ratings
“great” → +3.1

→

4

Add rules

“not good”
“VERY good!”

Crowdsourced ratings + researcher-designed rules

Hutto & Gilbert (2014), VADER, ICWSM

A dictionary built from human judgments — not a generative AI model.

How does a language model classify sentiment?

Two stages: learning patterns first, then applying them to new text

TRAINING

Large text datasets → learned language patterns

USING THE PRETRAINED MODEL

Input text

"The message is overwhelming."

→

Tokenize

The | message | is | overwhelming .

→

Attention

Context between tokens

→

Classify

Negative

Hugging Face Model Hub → Python / Colab → your text

We can load a pretrained classifier without training our own model.

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Choose a coding scheme BEFORE coding

OPERATIONAL DEFINITION

Positive

Mostly favorable evaluation or emotion toward the message.

Neutral / mixed

No clear direction, or substantial positive and negative elements.

Negative

Mostly unfavorable evaluation or emotion toward the message.

Different coding rules can yield different labels.

Human coding: make a judgment first

STUDENT ACTIVITY

1. Read the six sample responses in the notebook.

2. Assign each a Positive, Neutral/Mixed, or Negative label.

3. Write down one disagreement and why it happened.

VADER turns scores into categories

Compound score	Classroom label	Example
≥ +0.05	Positive	“This was helpful.”
Between −0.05 and +0.05	Neutral / mixed	“I saw the message.”
≤ −0.05	Negative	“This was disappointing.”

These conventional cutoffs are decisions, not universal truths.

Colab demonstration: VADER

STUDENT ACTIVITY

1. Upload the same dataset and run the VADER cells.
2. Compare the model labels to your group's original decisions.
3. Choose one example where the model label seems wrong or incomplete.

Emphasize checking original text before evaluating any label.

Two approaches to sentiment analysis

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Method B: a pretrained transformer classifier

COMPARE

What is different?

A model learns patterns from examples during training.

It predicts one of its trained sentiment labels.

What it is NOT

It is NOT the same as a chat-style generative LLM.

We do not prompt it to invent a new response.

Model hub = place to find and reuse existing models.

What happens behind a classifier?

HUGGING FACE

1 • Tokenize

The input text becomes tokens or token IDs.

2 • Model

The pretrained transformer processes contextual patterns.

3 • Predict

A classification head returns sentiment scores and a label.

Adapted conceptually from Hugging Face “Behind the pipeline.”

What does “confidence = 0.93” mean?

WORKED EXAMPLE

Model output: NEGATIVE, score 0.93

Strong model preference for that label—not 93% proven correctness.

Verify with human-labeled examples from the intended domain.

Live model demo

STUDENT ACTIVITY

1. Run the pretrained Hugging Face classifier on 18 sample responses.
2. Compare its predictions with VADER and human labels.
3. Identify a case where the model misses mixed sentiment or the target.

Model download requires internet and may take time.

Three approaches: what changes?

Human coders

Can reason about target and context; may disagree and need a codebook.

Lexicon rules

Transparent and fast; can miss context and subtle meaning.

Pretrained classifier

Context-sensitive pattern recognition; inherits training-domain limitations.

None automatically provides ground truth.

How do we validate automated labels?

COMPARE

Compare to human coding

Make a labeled sample with explicit rules.

Inspect agreement and disagreement cases.

Check challenging examples

Negation; sarcasm; mixed feelings; unclear targets; topic-specific terms.

Revise the codebook as needed.

Human judgment should be systematic—not just intuition.

Back to Block 3: what test fits?

CONNECT ACROSS BLOCKS

Predictor (X)	Outcome (Y)	Question	Candidate test
condition: categorical	sentiment label: categorical	Related proportions?	chi-square
condition: categorical	sentiment score: numeric	Different mean scores?	t-test*
message_trust: numeric	sentiment score: numeric	Associated scores?	correlation*

***Requires appropriate measurement assumptions; VADER score is not automatically interval-scale.**

What would chi-square tell us?

WORKED EXAMPLE

Condition × Positive / Neutral / Negative labels

Are the distributions of categories associated in this dataset?

It does not validate the labels or establish the meaning of a difference.

Could we ask a user-friendly generative LLM like ChatGPT instead?

COMPARE

Yes — as an extension

Prompt: “Classify sentiment toward the message as positive, neutral/mixed, or negative. Return one label.”

But document the method

Record the prompt, model, settings, codebook, human validation, and errors.

LLM agreement is not a substitute for validity.

What this tells us about communication research

KEY TAKEAWAYS

01 Open-ended responses can become research variables—but only after making coding decisions.

02 The tool choice depends on the construct, domain, and validation evidence.

03 Statistical inference comes after measurement quality, not before it.

Researcher → operationalization → model evaluation → interpretation.

Exit ticket: show what you learned

EXIT TICKET

1. How is sentiment classification different from counting words?

2. Give one reason a sentiment classifier could mislabel a response.

3. Which Block 3 test fits condition × sentiment category, and why?

Assignment 5 — Final Project Analysis Plan

FINAL PROJECT CHECKPOINT

Due: Thursday, December 3, 2026

During the final-project workshop class • 10% of course grade

Before the workshop

- Form a group of approximately 4–5.
- Choose an instructor-provided instructional dataset and research question.
- Prepare to explain your analysis plan.

During your group consultation

- Research question and relevant variables
- Proposed method and why it fits
- Expected analysis workflow
- Potential challenges

No separate written report required unless requested. Goal: refine your plan before the final analysis.